United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Exam Name: Mid Term Exam   Trimester: Fall 2025
Course Code: PMG 4101, Course Title: Project Management
Total Marks: **30**   Duration: 1 hour 30 minutes

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

Answer all the questions.

Consider the following Scenario:

A private space exploration company has tasked your team with developing a Virtual Reality (VR)-Driven Remote Collaboration Platform to support engineers, scientists, and astronauts on space missions. The platform aims to create immersive, real-time virtual environments where teams on Earth and in space can collaborate seamlessly. It will simulate spacecraft interiors, space stations, and mission operations, allowing users to interact with 3D models, perform assembly tasks, troubleshoot systems, and rehearse spacewalks and emergency protocols.

The platform will include key features like real-time collaboration, AI-powered troubleshooting, and astronaut training modules that simulate zero-gravity environments. Engineers on Earth will assist astronauts through mixed-reality systems, and the platform will provide real-time data visualizations for monitoring life support, spacecraft health, and experiments. The system must also integrate with existing spacecraft systems and handle deep-space communication challenges, such as network latency and bandwidth limitations.

User experience is crucial, with a focus on ensuring comfort in virtual environments, especially during zero-gravity simulations, and accurately modeling space hazards like space debris and radiation. Cybersecurity will be a priority, as the platform will handle sensitive astronaut and mission data. The platform's AI capabilities will predict mission outcomes, monitor system health, and provide real-time troubleshooting assistance. Development will be phased, starting with astronaut training and real-time collaboration, followed by advanced AI and system monitoring features.

This project will require collaboration among space engineers, VR developers, AI specialists, and cybersecurity experts to create a reliable, scalable platform that supports mission planning, training, and execution.

1. Choose a Software Engineering ***Methodology*** for developing the above-mentioned project scenario and discuss the rationale for choosing that methodology. Describe the roles and responsibilities of the ***Product Owner*** and ***Scrum Master*** in the Scrum process.

[CO1]   3+2

2. Define ***"Statement of Work"*** in project planning and mention its elements. Write down the **Vision** and **Scope** Document for the project described above.

[**CO2**] **1+2+2**

3. Mention the ***Major Principals*** followed in WBS estimation method. Develop a ***"Work Breakdown Structure"*** for the above mention scenario following the principles/rules.

[CO3] 1+4

4. Using a diagram, show each stage in the ***"Basic Course of Events"*** for the Delphi Wideband Estimation method. Suppose you want to form an estimation team consisting of **five members**. Show the steps of estimation session (one estimation form and tentative convergence chart of duration) for the above-described project, according to DWE method. [**CO3**] **2+3**

5. Give a brief explanation of the ***RMMM*** concept. Now as a project manager you are preparing for Risk Mitigation. Develop a ***Risk Plan Sheet*** showing at least three major risks for the above scenario. **[CO2] 2+3**

6. Progress report submission and presentation on Work Breakdown Structure and Delphi Wideband Estimation Technique for a group project have been evaluated earlier (**Do not answer this question**, **already completed in classroom**). **[CO3] 5**